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Ex 16: Feed Forward Neural Network

CODE:

```
import numpy as np
def sigmoid(x):
  return 1 / (1 + np.exp(-x))
def sigmoid_deriv(x):
  return x * (1 - x)
# XOR Inputs and Expected Outputs
X = np.array([[0,0], [0,1], [1,0], [1,1]])
y = np.array([[0], [1], [1], [0]])
np.random.seed(42)
w1 = np.random.uniform(size=(2, 4))
w2 = np.random.uniform(size=(4, 1))
for _ in range(10000):
  # Forward Pass
  h = sigmoid(np.dot(X, w1))
  out = sigmoid(np.dot(h, w2))
  # Backward Pass
  error = y - out
  d_out = error * sigmoid_deriv(out)
  d_h = d_out.dot(w2.T) * sigmoid_deriv(h)
  w2 += h.T.dot(d_out) * 0.1
  w1 += X.T.dot(d_h) * 0.1
```

```
print("Predictions after training:")

print(np.round(out, 2))
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-16.py ==========
Predictions after training:
[[0.1 ]
 [0.93]
 [0.92]
 [0.06]]
```